



deti

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departamento de eletrónica,
telecomunicações e informática

Real-time Weapon Detection in Surveillance Video Footages

Mestrado em Engenharia Informática

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Introduction

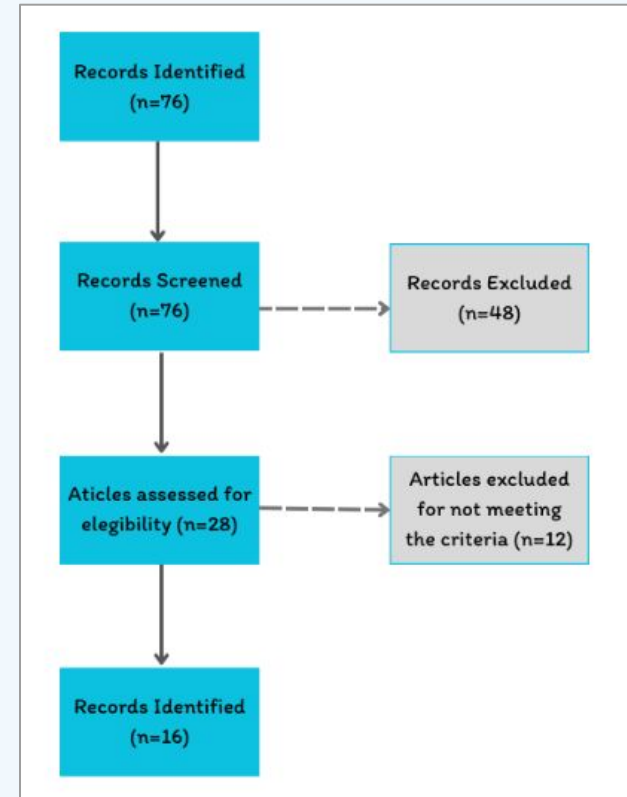
- Rising crime rates despite strict weapon laws
- Increased reliance on CCTV systems
- Limitation of human-monitored CCTV systems
- **Goal:** Automated weapon detection system

Participações registadas por distrito e região autónoma												
Distritos		Ano 2021	Ano 2022	Var %	Dif		Distritos		Ano 2021	Ano 2022	Var %	Dif
✖	Aveiro	400	470	17,5 %	70	!	Portalegre	93	93	0,0 %	0	
✖	Beja	84	120	42,9 %	36	✖	Porto	1 886	2 100	11,3 %	214	
✔	Braga	469	445	-5,1 %	-24	✖	Santarém	314	375	19,4 %	61	
✔	Bragança	56	46	-17,9 %	-10	✖	Setúbal	1 234	1 330	7,8 %	96	
✖	Castelo Branco	106	148	39,6 %	42	✖	Viana do Castelo	102	124	21,6 %	22	
✖	Coimbra	252	317	25,8 %	65	✔	Vila Real	100	90	-10,0 %	-10	
✖	Évora	93	136	46,2 %	43	✖	Viseu	153	158	3,3 %	5	
✖	Faro	707	876	23,9 %	169	✖	Madeira	177	269	52,0 %	92	
✖	Guarda	56	60	7,1 %	4	✖	Açores	220	232	5,5 %	12	
✖	Leiria	286	344	20,3 %	58	✖	Não especificado	207	402	94,2 %	195	
✖	Lisboa	4 619	5 146	11,4 %	527	✖	Total	11 614	13 281	14,4 %	1 667	

Geographical Distribution of Crime Rate Increase in Portugal by District in 2022, from RASI

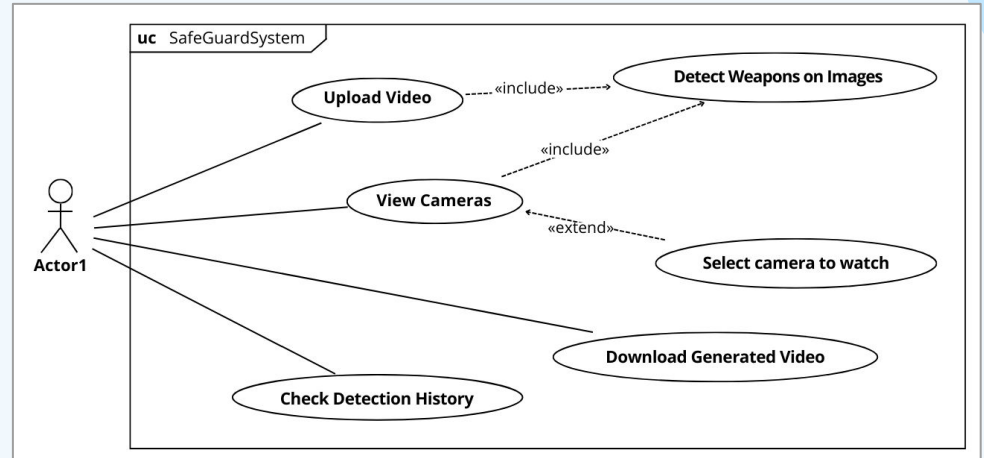
Related Work

- Search applied over the following research platforms
 - ResearchGate
 - Scopus
 - arXiv
- Strategic keywords selection
- Emphasis on recent publications
- Example query:
 - ("Deep" AND "Learning" AND "Real" AND "Time" AND "Weapon" AND "Detection")

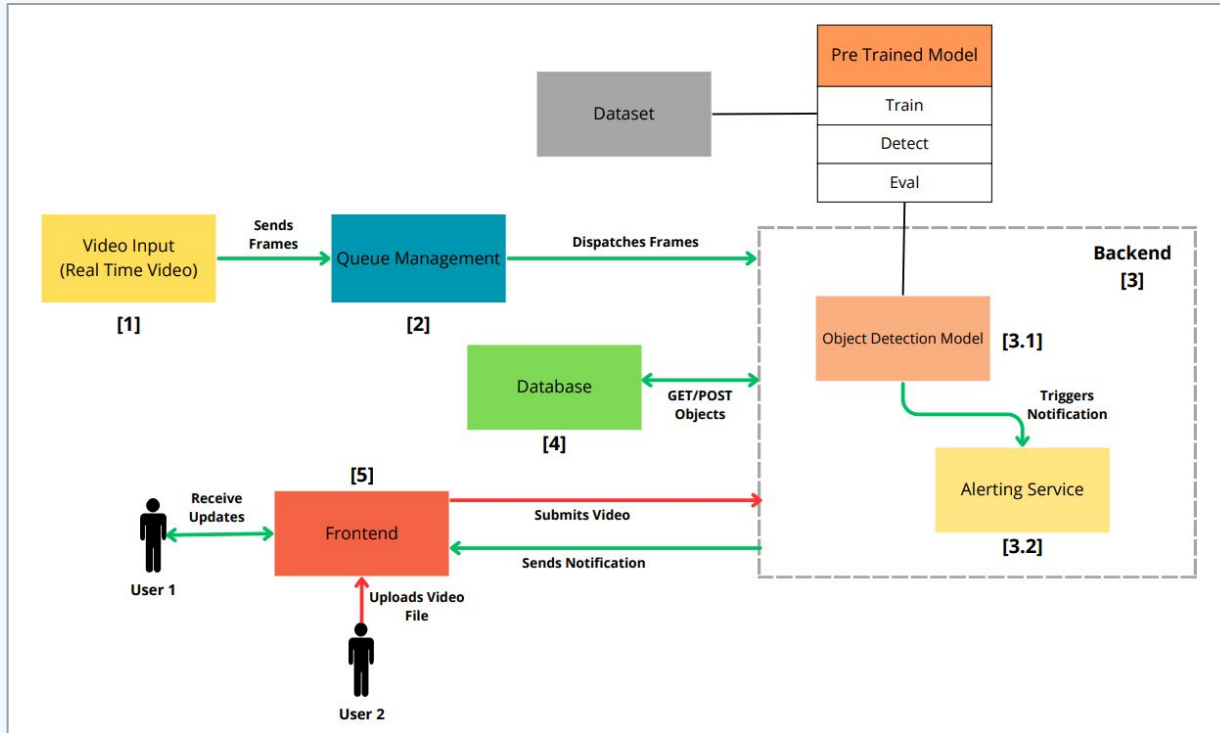


System Use Cases

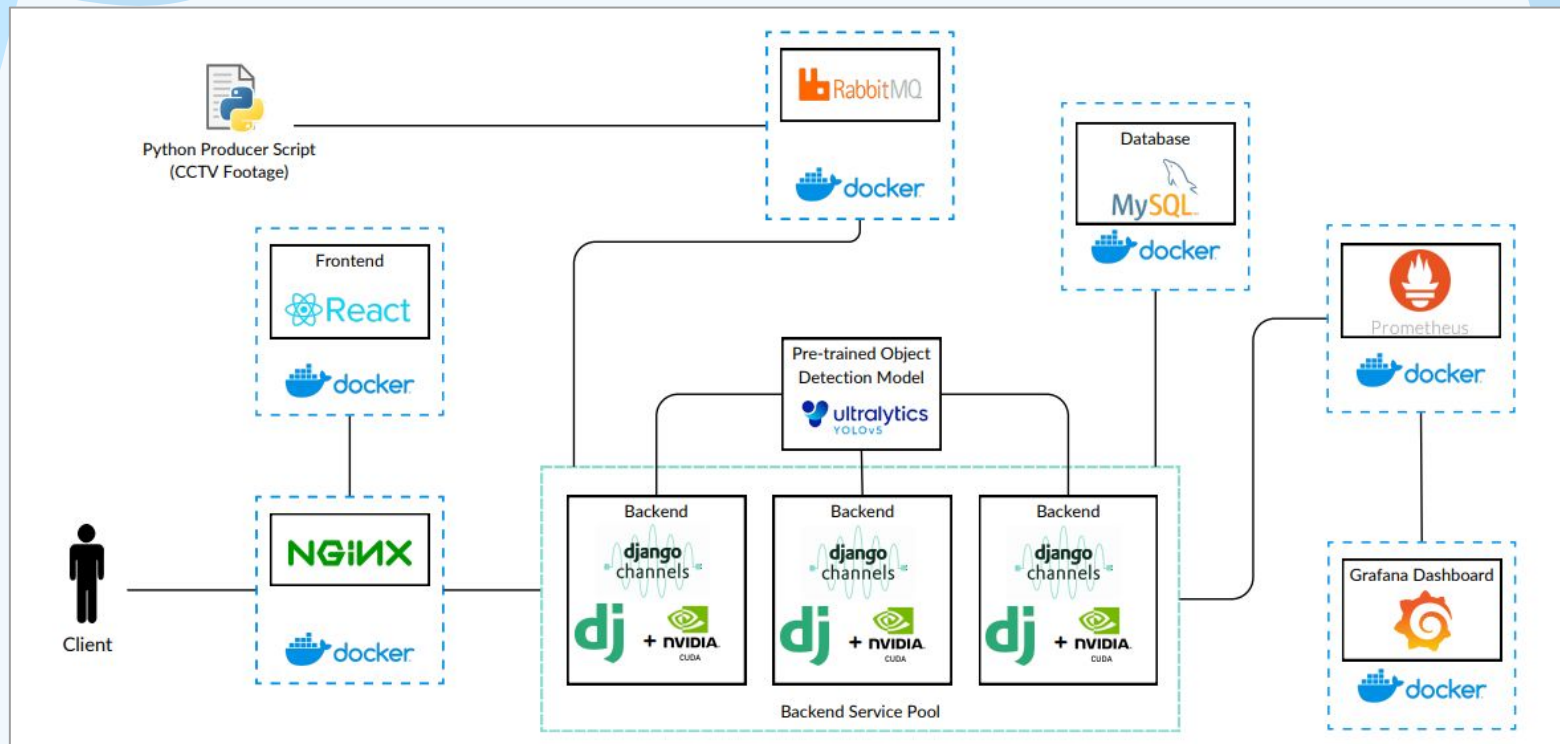
- Upload pre-recorded video
- View listed cameras
- Select camera to monitor
- Download model generated video
- Check all detections history



High Level Architecture



Implementation Architecture



Dataset Sources

- DaSCI
- Mock Attack Dataset
- Action Recognition and Object Detection

Available at: <https://github.com/pedromonteiro01/FirearmsKnivesDataset>

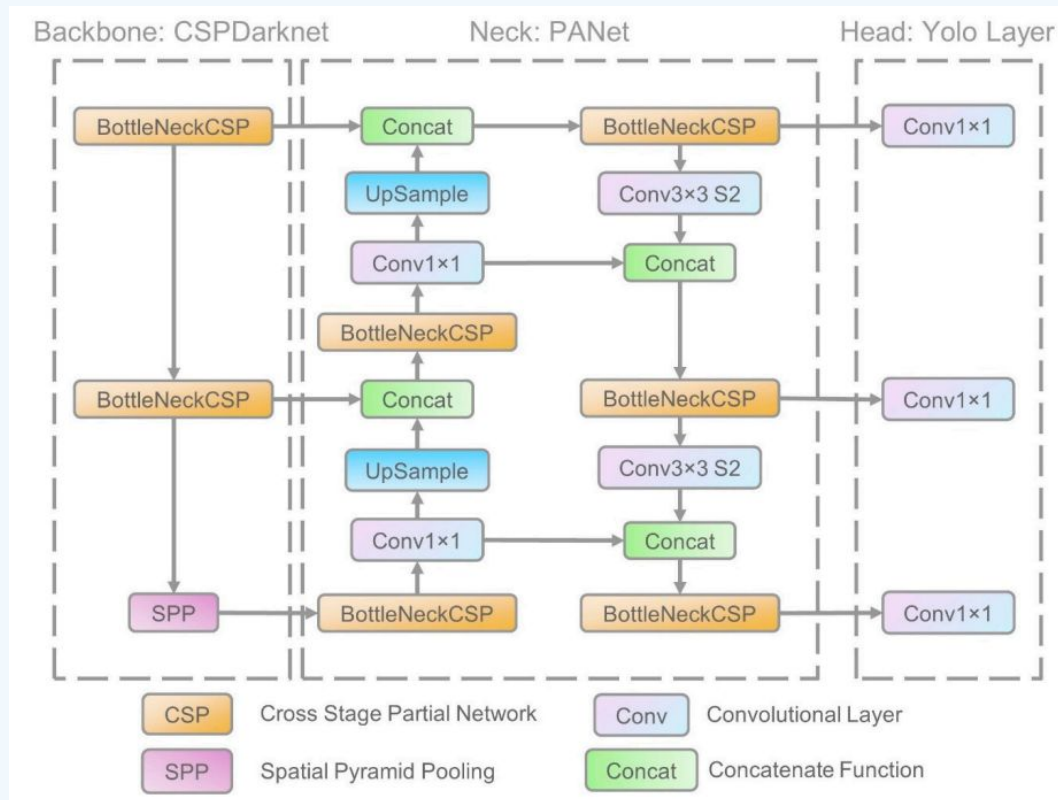


Source	Images	Brief Description
DaSCI	5149	Variety of weapons displayed in different positions and random everyday objects.
Mock Attack	2895	Two cameras located in corridors, each with uniform lighting and various obstacles like doors or bins. Another camera is positioned at the entrance of a university building, featuring a black carpet and uneven lighting.
Action Recognition and Object Detection	2976	Multiple videos that feature three categories: handguns, machine guns, and no guns (with objects easily confused with weapons). Each video is set in various scenarios, providing diverse backgrounds and contexts for each category.

YOLOv5 Model Overview

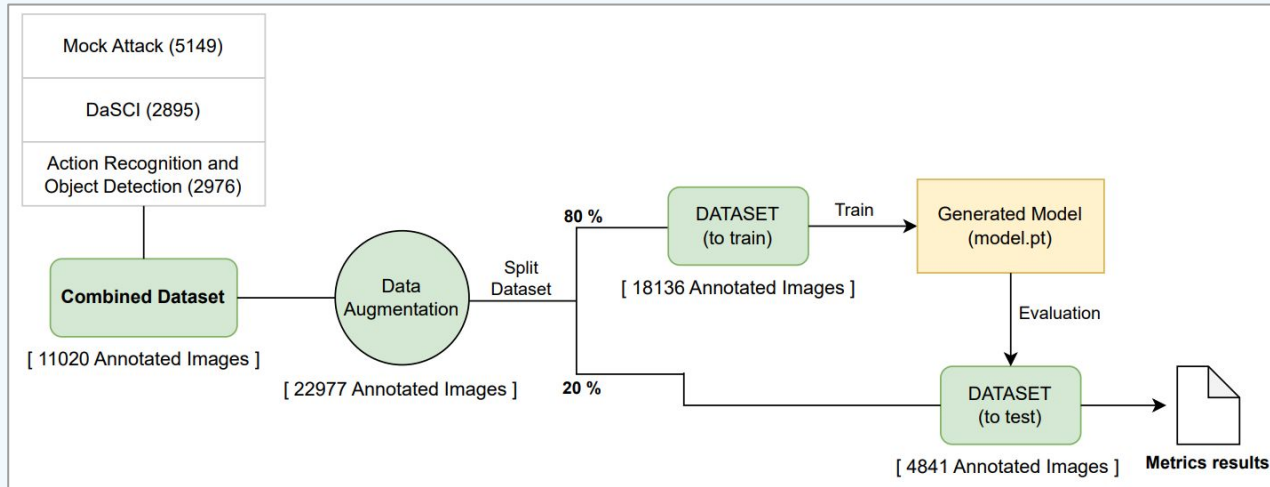
Three primary sections:

- Backbone
- Neck
- Head



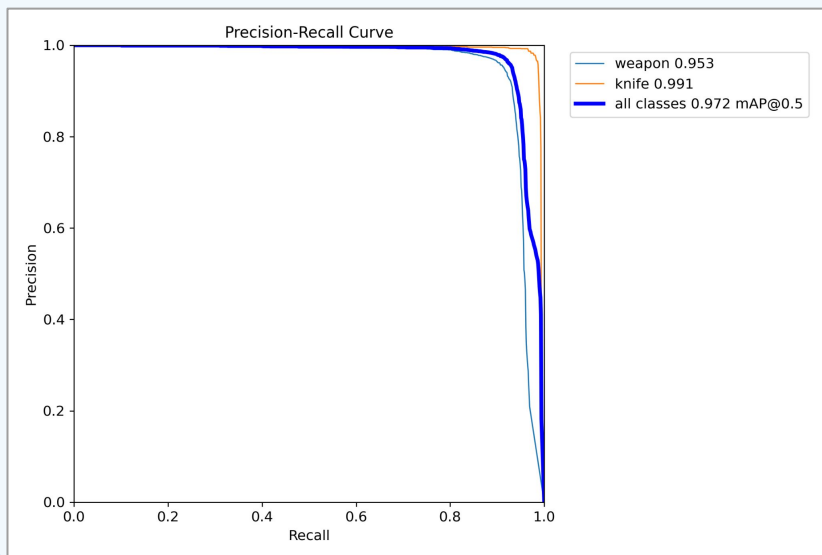
Object Detection Model Creation

- Data Augmentation
 - 1st phase: rotations and blurr
 - 2nd phase: brightness changes
- 80/20 split ratio

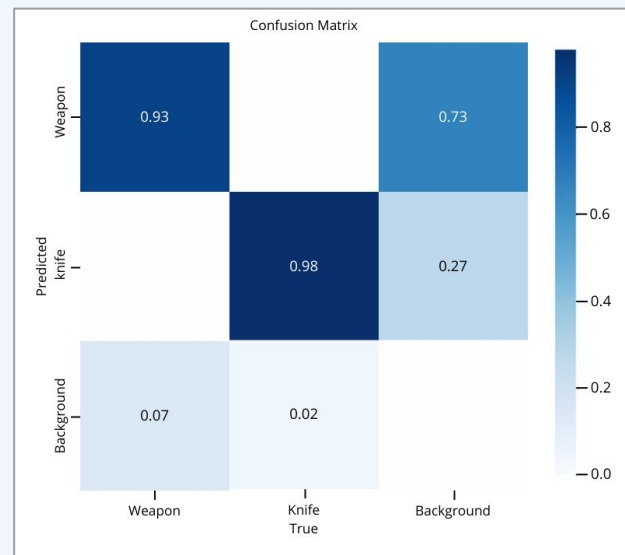


YOLOv5 Model Results

Epochs	Batch Size	Image Size	Accuracy	Precision	Recall
100	32	640	0.973	0.966	0.944

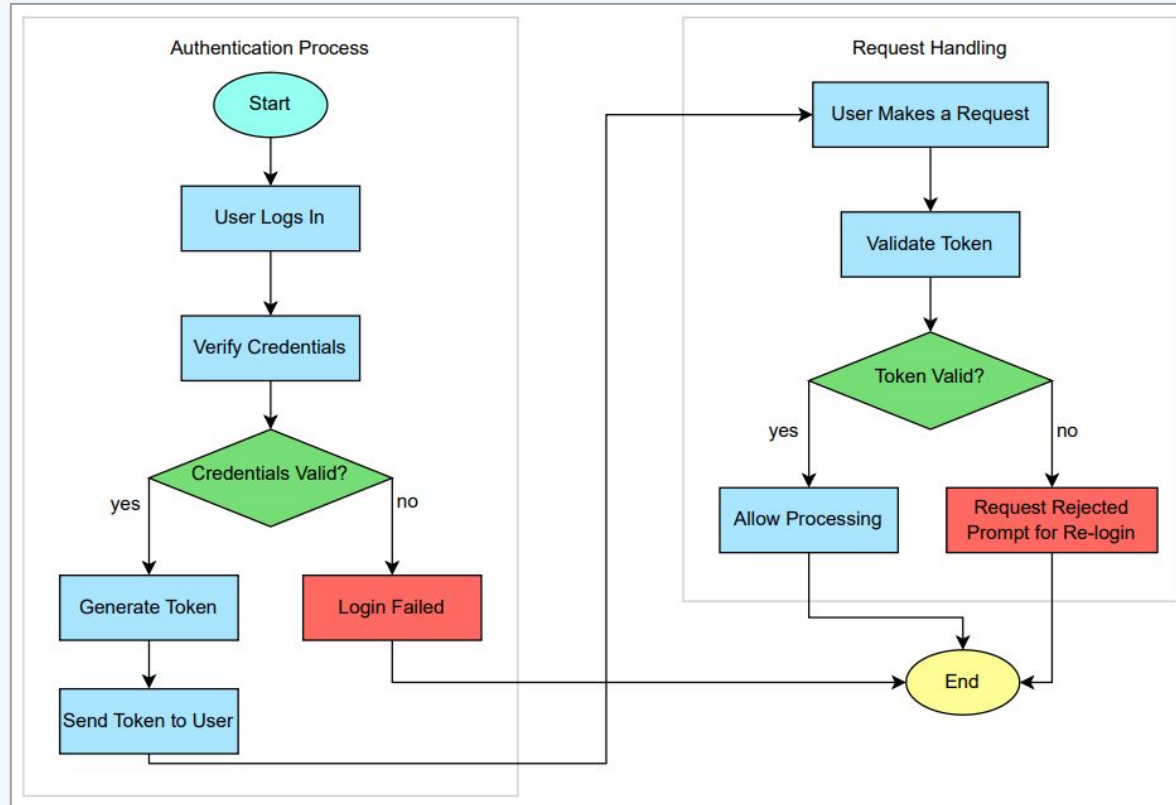


Precision Recall Curve

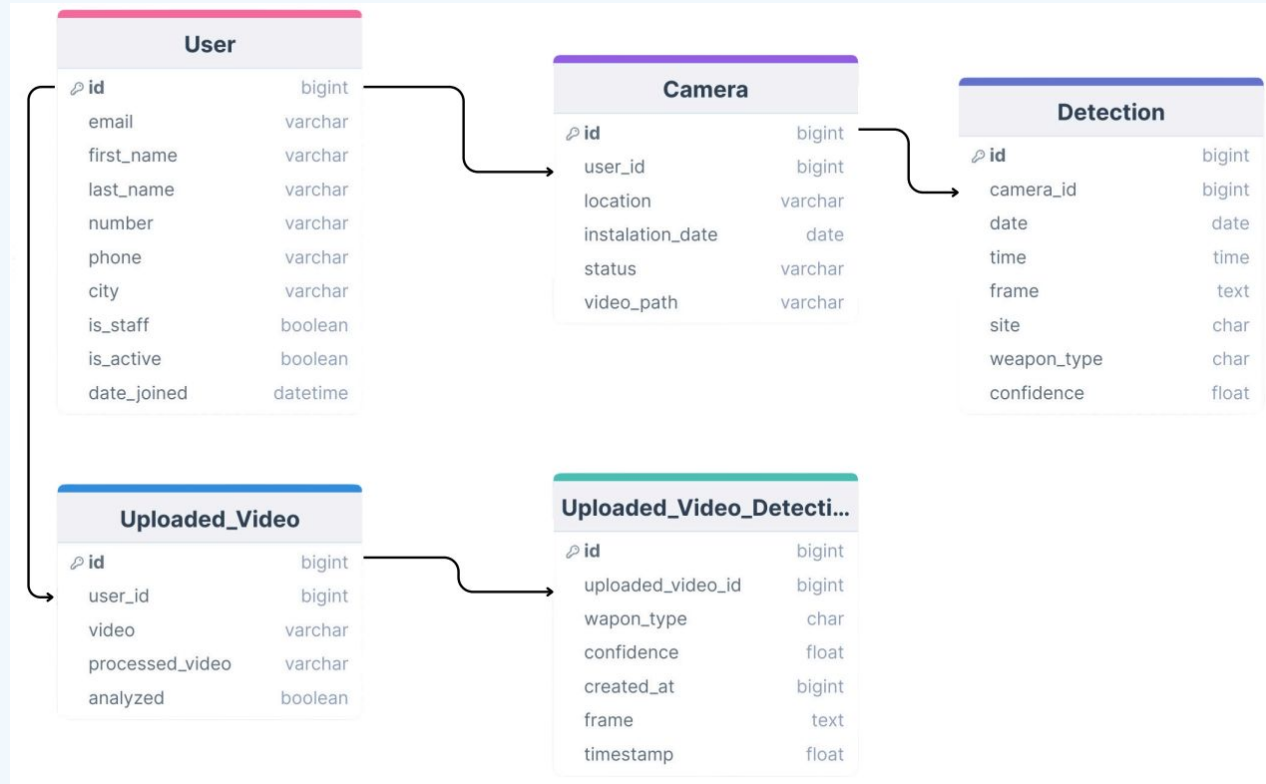


Confusion Matrix

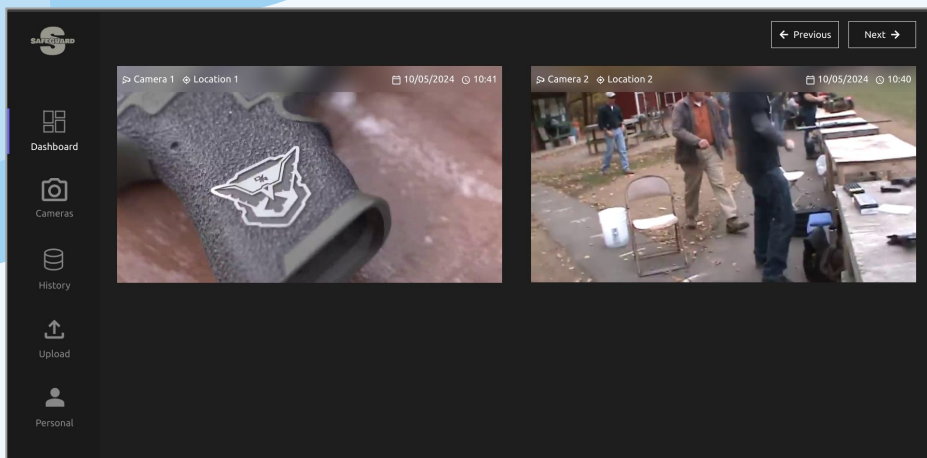
Authentication



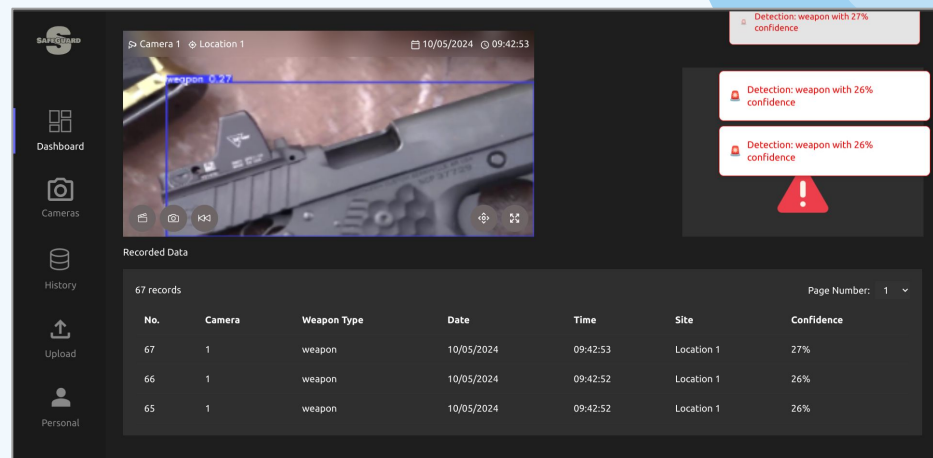
Database Schema



Web Application

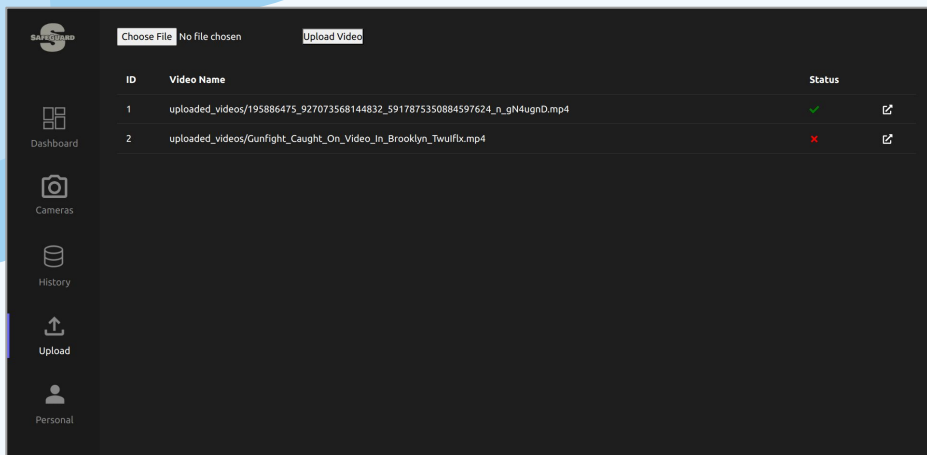


Multi-camera page

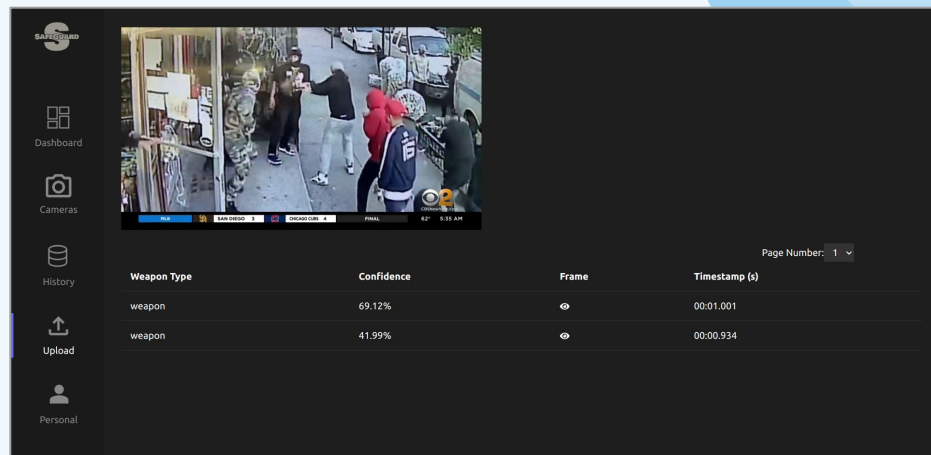


Single-camera analysis

Web Application

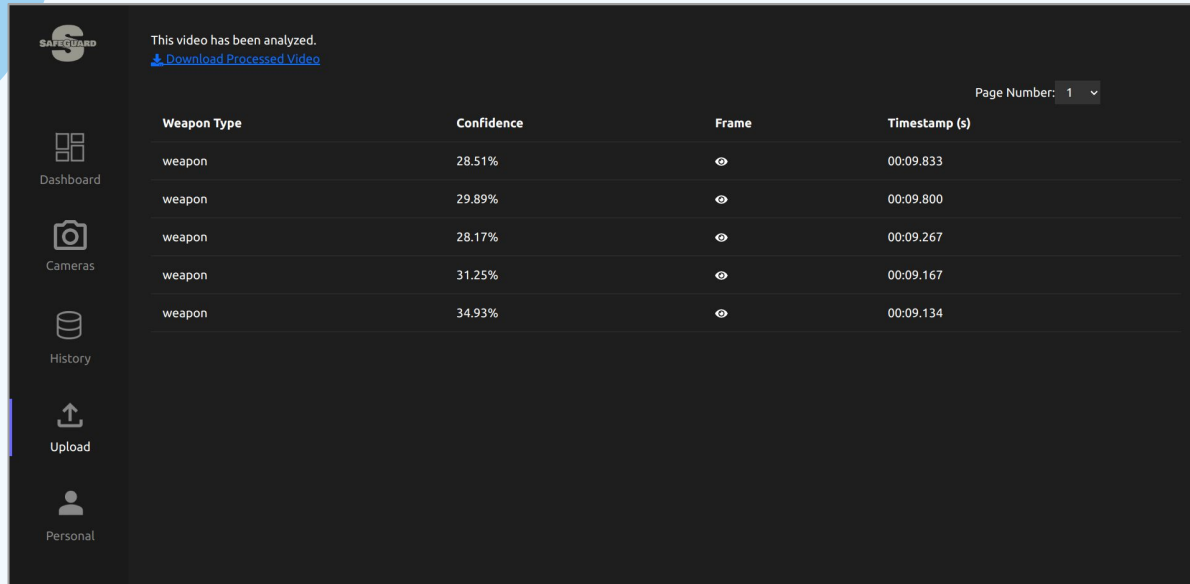


Upload video



Analysing uploaded video

Web Application



This video has been analyzed.
[Download Processed Video](#)

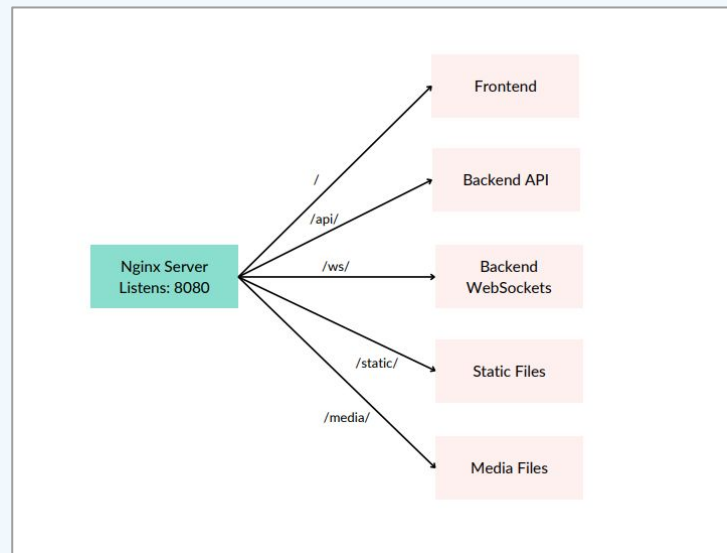
Page Number: 1 ▾

Weapon Type	Confidence	Frame	Timestamp (s)
weapon	28.51%	👁	00:09.833
weapon	29.89%	👁	00:09.800
weapon	28.17%	👁	00:09.267
weapon	31.25%	👁	00:09.167
weapon	34.93%	👁	00:09.134

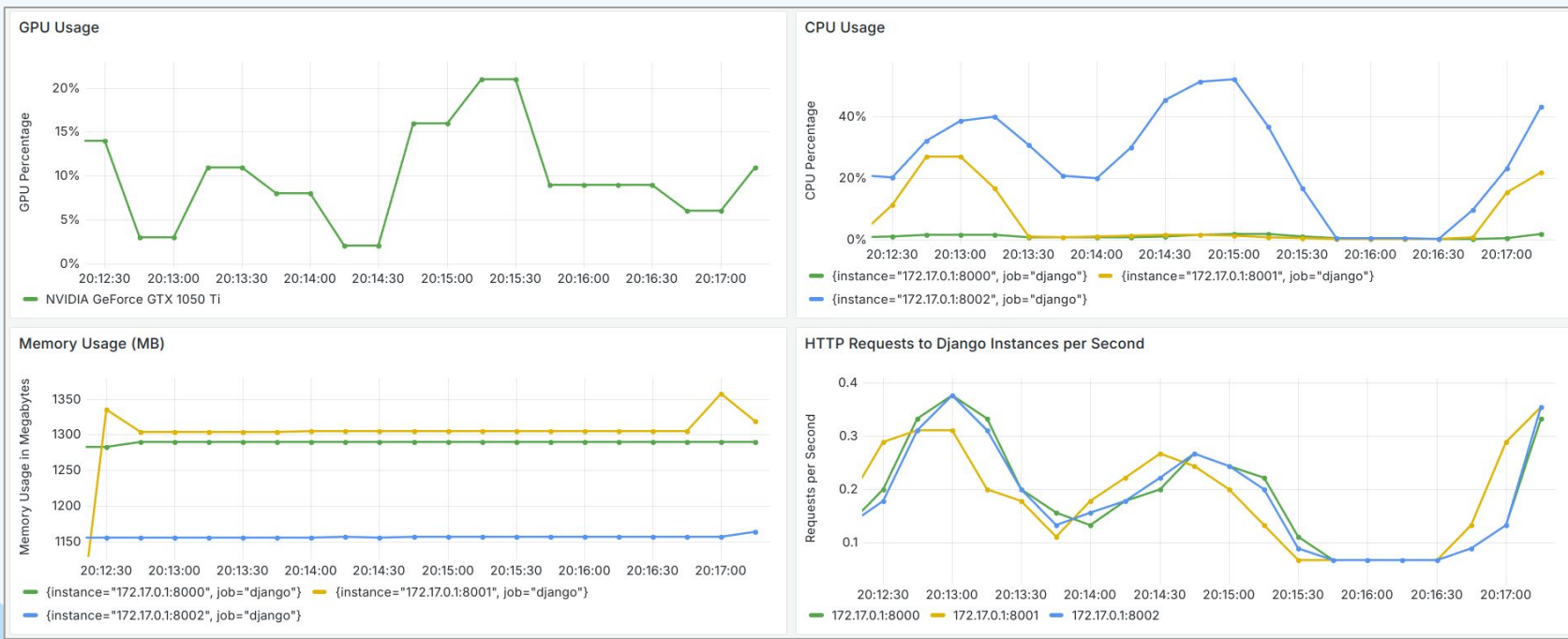
Results from uploaded video

Nginx Load Balancing

- High-performance web server and reverse proxy
- Manages incoming HTTP and WebSocket requests
- Enhances the system's reliability and scalability



Performance Monitoring

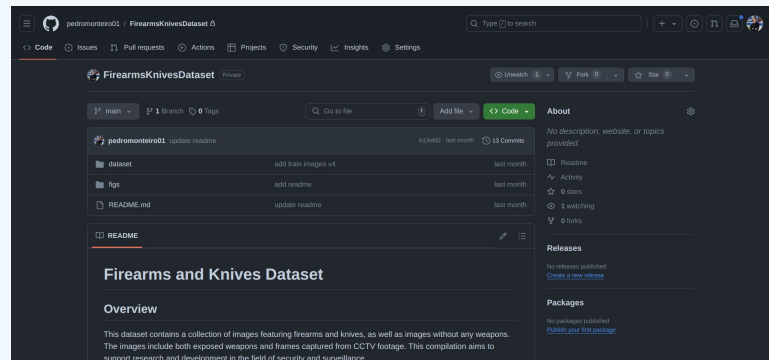


Scalability and Performance Optimizations

- Web Workers to handle frame processing in the background
- Reduce number of DOM updates in canvas
- Use single RabbitMQ persistent connection
- Reduce frequency of frames analyzed (each 10th frame)
- Frames batch processing
- Optimize RabbitMQ settings

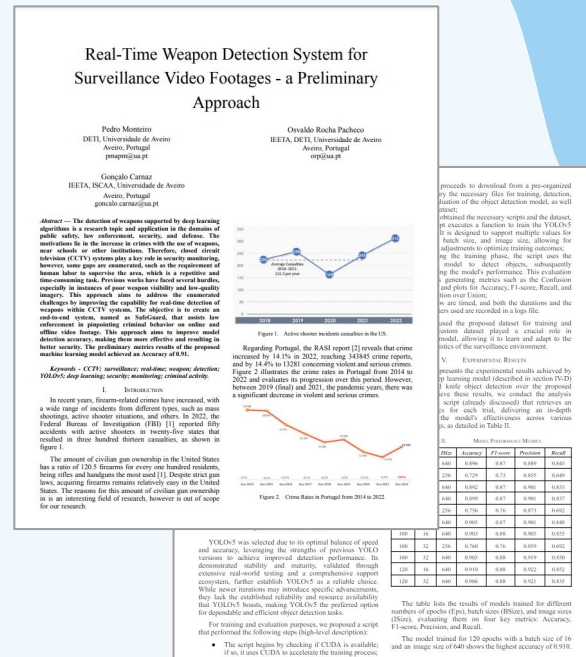
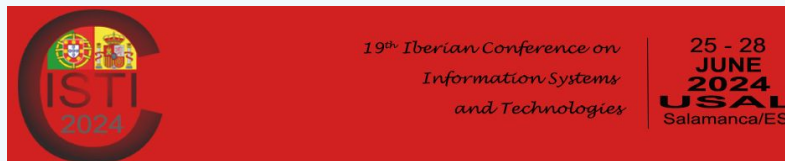
Results

- System prototype (integrates a fine-tuned weapon detection YOLOv5 model)
- Compilation and publication of a fully annotated dataset
- Scientific publication presented on CISTI'2024
- E-poster published on “Encontro Ciência 2024”



Scientific Publication

- CISTI'2024 - 19th Iberian Conference on Information Systems and Technologies
- June 25th - 28th, 2024
- First design of the application and initial results of the object detection model



E-poster

- Encontro Ciência 2024, Porto
- July 3rd - 5th, 2024
- Explore thesis objectives and contextual background, while also detailing the application illustrating its functionality, architecture, and outcomes



SafeGuard: A Real-time Weapon Detection System.

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Introduction

An increase in firearm-related crimes has been observed in recent years, as evidenced by the RASD report [1]. This trend is evident in several Portuguese districts, such as Lisbon (+11.4%), Porto (+11.3%), and Faro (+23.9%). This has led to the emergence of a pressing challenge for law enforcement agencies and the academic community (as a scientific partner), namely the prevention of firearms.

The SafeGuard system, depicted in Figure 1, operates by analyzing video frames from CCTV footage in real-time. When potential threats of firearm-related activities are detected, the system generates alerts, which are promptly displayed on the web interface. Police officers using the application can then rapidly respond to these alerts for swift intervention.

Fig. 1. SafeGuard Application Overview.

Objectives

The main objectives can be described as:

- Conduct a thorough analysis of existing research, algorithms, architectures, and datasets in weapon detection;
- Develop a scalable system, SafeGuard, that handles multiple cameras simultaneously, with real-time alerting mechanisms for weapon detection;
- Build a comprehensive and diverse dataset with images and video footage of various weapon types in different environments and conditions.

Methodology

To implement a system capable of detecting weapons, we established the following methodology:

- State of the art review;
- Propose a high-level architecture;
- Establish the architecture with the associated technologies (to resolve the constraints associated with these applications, such as data volume, real-time constraints, and others);
- Propose a detection model using YOLOv5. For this a combined dataset that uses available datasets with some adjustments was developed;
- Application implementation from scratch, and model performing tests.

The second phase introduced variations in lighting conditions. These augmentations increased the diversity of the dataset. Following augmentation, the dataset was split into training and testing sets using an 80/20 ratio, as illustrated in Figure 3.

Fig. 3. Model Creation Steps

The architecture components are outlined in the Table 1:

Table 1. System Components and Descriptions

Component	Description
Python Producer Script	Sends simulated footage to RabbitMQ.
RabbitMQ	Ensures smooth and reliable data handling as a message broker.
Image Extractor	Utilizes pre-trained YOLOv5 model to process incoming footage. Uses Django Channels for real-time communication and immediate alarm generation based on detected objects.
React Frontend	Presents a user-friendly web interface. Allows users to interact with the system, view real-time footage, upload files to analyze.
Load Balancing with Nginx	Manages incoming traffic efficiently. Distributes load across multiple servers. Ensures high availability and scalability of the application.
Performance Monitoring	Prophetiler used for real-time performance monitoring. Grafana offers a detailed dashboard for monitoring metrics.

Tests and Results

The YOLOv5 model was trained under various configurations, adjusting parameters such as the number of epochs, batch size, and image size to find the optimal performance.

After extensive testing and parameter adjustments, the model optimal performance is represented in Table 2, reaching a maximum accuracy of 0.973.

Table 2. Optimal Configuration Parameters and Results

Epochs	Batch Size	Image Size	Accuracy	Precision	Recall
100	32	640	0.973	0.969	0.948

Figure 4 exhibits an output from one dataset frame, processed using the fine-tuned YOLOv5 model. The model has annotated four objects with bounding boxes, identifying them as potential weapons with varying degrees of confidence: one at 0.61, and two others at 0.65 and 0.75, and a knife at a confidence of 0.55.

Conclusions

- Development of a real-time weapon detection and alarm system
- Dataset definition and model training
- Pretrained model integration into a web application
- Challenges and considerations
 - Difficulties in specific scenarios
 - Challenges with real-time processing

Future Work

- Auto-Scaling Mechanisms
- Upgrade System Hardware
- Exploration of newer versions of the YOLO algorithm
- Dataset Adaptation



Demo



Thanks

Questions?

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